

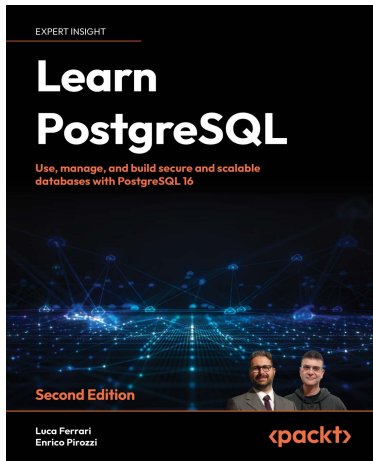
Databases

SQL Functions.

Ferrari & Pirozzi

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Databases: SQL Functions.



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Overview

- ▶ PostgreSQL supports server-side programming via functions.
- ▶ Built-in languages: SQL, PL/pgSQL, C.
- ▶ Optional: PL/Python, PL/Perl, PL/Java, etc.
- ▶ This chapter focuses on SQL and PL/pgSQL functions.

The Function Syntax

```
1 CREATE FUNCTION function_name(p1 type, p2 type, p3 type, ..., pn type)
2 RETURNS type AS
3 BEGIN
4   -- function logic
5 END;
6 LANGUAGE language_name
```

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4. For the PL/pgSQL language, put some code between the **BEGIN** and **END** blocks.
5. For the PL/pgSQL language, the function has to end with the **END** keyword followed by a semicolon.
6. Define the language in which the function was written (for example, sql or plpgsql, plperl, plpython, and so on).

Basic SQL Function Example

```
1 CREATE OR REPLACE FUNCTION my_sum(x integer, y integer)
2 RETURNS integer AS
3 $$
4     SELECT x + y;
5 $$ LANGUAGE SQL;
```

Call: `SELECT my_sum(1, 2);`

Function Returning a Set

Returns a set of primary keys of deleted records.

```
1 CREATE OR REPLACE FUNCTION delete_posts(p_title text)
2 RETURNS SETOF integer AS
3 $$
4     DELETE FROM posts WHERE title = p_title
5     RETURNING pk;
6 $$ LANGUAGE SQL;
```

Function Returning a Table

```
1 CREATE OR REPLACE FUNCTION delete_posts_table(p_title text)
2 RETURNS TABLE (ret_key integer, ret_title text) AS
3 $$
4     DELETE FROM posts WHERE title = p_title
5     RETURNING pk, title;
6 $$ LANGUAGE SQL;
```

Polymorphic SQL Function

- ▶ Polymorphic functions are useful for DBAs when we need to write a function that has to work with different types of data.
- ▶ We want to create a function that accepts two parameters and replaces the first parameter with the second one if the first parameter is NULL (Oracle NVL or PostgreSQL Coalesce).
- ▶ The problem is that we want to write a single function that is valid for all types of data (integer, real, text, and so on).

Polymorphic SQL Function

```
1 CREATE OR REPLACE FUNCTION nvl(anyelement, anyelement)
2 RETURNS anyelement AS
3 $$
4     SELECT COALESCE($1, $2);
5 $$ LANGUAGE SQL;
```

Works with multiple data types.

PL/pgSQL Function Structure

- ▶ The PL/pgSQL language is the default built-in procedural language for PostgreSQL.
- ▶ It can do the following:
 - ▶ Can be used to create functions and trigger procedures.
 - ▶ Add new control structures.
 - ▶ Add new data types to the SQL language.
- ▶ It supports the following:
 - ▶ Variable declarations.
 - ▶ Expressions.
 - ▶ Control structures as conditional structures or loop structures.
 - ▶ Cursors.

PL/pgSQL Function Structure

```
1 CREATE FUNCTION my_sum(x integer, y integer)
2 RETURNS integer AS
3 $$
4     DECLARE
5         ret integer;
6     BEGIN
7         ret := x + y;
8         RETURN ret;
9     END;
10 $$ LANGUAGE 'plpgsql';
```

Using IN/OUT Parameters

```
1 CREATE FUNCTION my_sum_3_params(IN x integer, IN y integer, OUT z integer) AS
2 $$
3 BEGIN
4     z := x + y;
5 END;
6 $$ LANGUAGE plpgsql;
```

Using IN/OUT Parameters

```
1 CREATE OR REPLACE FUNCTION my_sum_mul(IN x integer, IN y integer, OUT w integer, OUT
  ↪ z integer) AS
2 $$
3 BEGIN
4     z := x + y;
5     w := x * y;
6 END;
7 $$
8 language 'plpgsql';
```

Function Volatility Categories

- ▶ **VOLATILE** – default; can modify the database; result can change.
- ▶ **STABLE** – cannot modify the database; same result for same input in a transaction.
- ▶ **IMMUTABLE** – cannot modify the database; result is constant forever for same input.

Conditional Logic - IF Statement

```
1 IF x > y THEN
2     RETURN 'x > y';
3 ELSIF x < y THEN
4     RETURN 'x < y';
5 ELSE
6     RETURN 'x = y';
7 END IF;
```

Conditional Logic - IF Statement

```
1 CREATE OR REPLACE FUNCTION my_check(x integer default 0, y integer default 0)
  ↳ RETURNS text AS
2 $BODY$
3 BEGIN
4     IF x > y THEN
5         return 'first parameter is greater than second parameter';
6     ELSIF x < y THEN
7         return 'second parameter is greater than first parameter';
8     ELSE
9         return 'the 2 parameters are equals';
10    END IF;
11 END;
12 $BODY$
13 language 'plpgsql';
```

Loop Example with Composite Return

```
1 CREATE TYPE my_ret_type AS (  
2   id integer, title text, record_data hstore  
3 );  
4  
5 CREATE FUNCTION my_first_fun(p_id integer)  
6 RETURNS SETOF my_ret_type AS  
7 $$  
8   DECLARE  
9     rw posts%ROWTYPE;  
10    ret my_ret_type;  
11  BEGIN  
12    FOR rw IN SELECT * FROM posts WHERE pk = p_id LOOP  
13      ret.id := rw.pk;  
14      ret.title := rw.title;  
15      ret.record_data := hstore(  
16        ARRAY['title', rw.title, 'Title and Content', format('%s %s', rw.title, rw.content)]  
17      );  
18      RETURN NEXT ret;  
19    END LOOP;  
20    RETURN;  
21  END;  
22 $$  
23 LANGUAGE 'plpgsql';
```

Exception Handling

```
1 CREATE FUNCTION my_second_except(x real, y real)
2 RETURNS real AS
3 $$
4 DECLARE
5     ret real;
6 BEGIN
7     ret := x / y;
8     RETURN ret;
9 EXCEPTION
10    WHEN division_by_zero THEN
11        RAISE INFO 'DIVISION BY ZERO';
12        RAISE INFO 'Error % %', SQLSTATE, SQLERRM;
13    RETURN 0;
14 END;
15 $$
16 LANGUAGE 'plpgsql';
```



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- 3 Functions in PostgreSQL can return more than just single values; they can output entire result sets or tables using `RETURNS SETOF` or `RETURNS TABLE`, which can then be queried like a standard table.

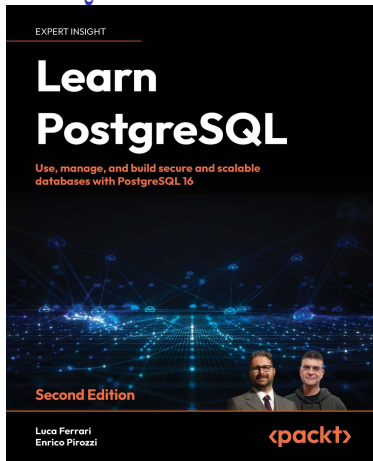
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- 1 The **SECURITY DEFINER** clause is a key security feature that allows a function to execute with the permissions of its owner rather than the calling user, enabling controlled access to restricted data.

Additional Resources

1. “*PostgreSQL Documentation - User-Defined Functions*” by The PostgreSQL Global Development Group. Available at <https://www.postgresql.org/docs/current/xfunc.html>.
2. “*How to write SQL Functions in PostgreSQL FAST — Full Tutorial — Step-by-Step*” by Software Nuggets. Available at <https://www.youtube.com/watch?v=GNTap3iVg3c>.
3. “*Como crear una función (Stored Procedure) - FUNCTION — PostgreSQL #31*” by TecnoBinaria. Available at <https://www.youtube.com/watch?v=cQPcqIkBV-U>.

Database Administration: SQL Functions.



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